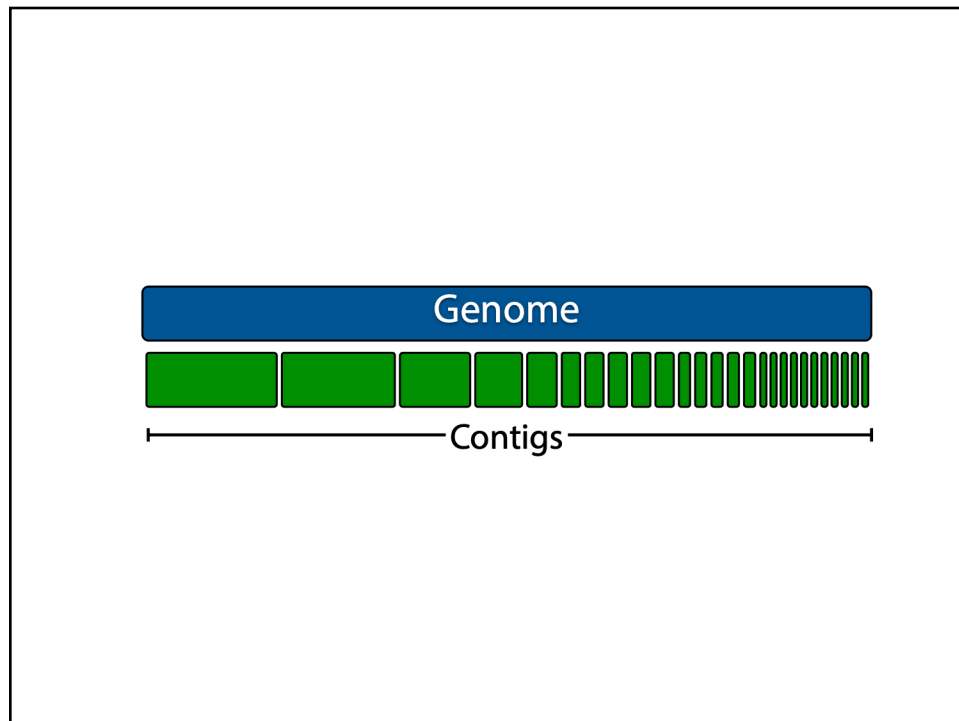




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N50 size

Def: 50% of the genome is in contigs as large as the N50 value

Example: 1 Mbp genome

50%



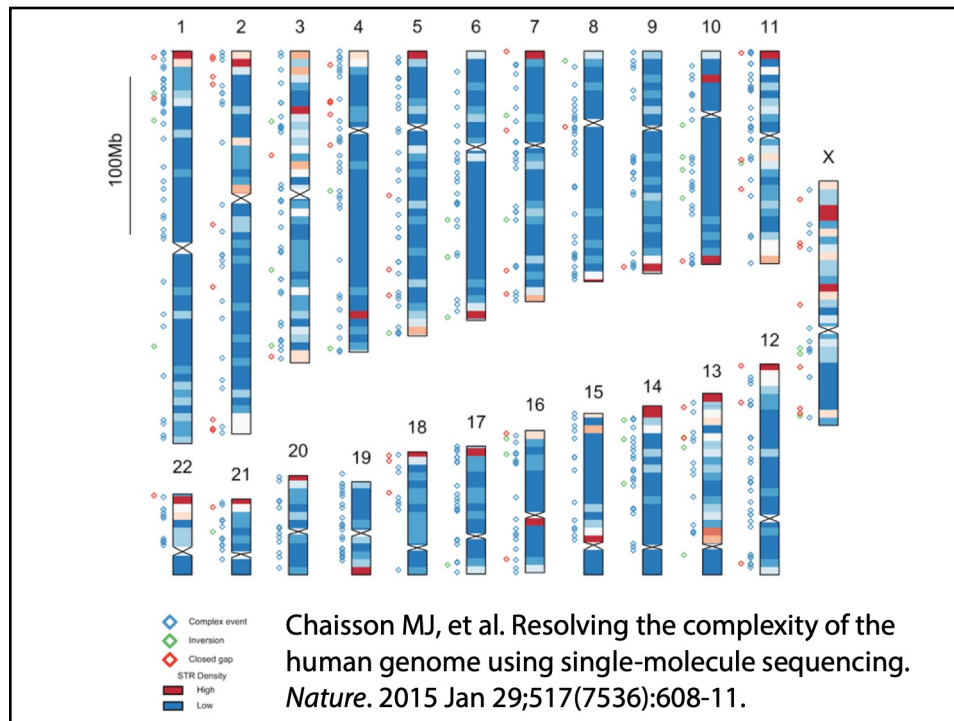
N50 size = 30 kbp

$(300k + 100k + 45k + 45k + 30k = 520k \geq 500kbp)$

Note:

N50 values are only meaningful to compare when base genome size is the same in all cases

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Scaffolding

Both OLC and DBG are concerned with constructing the longest, most accurate *contigs* possible

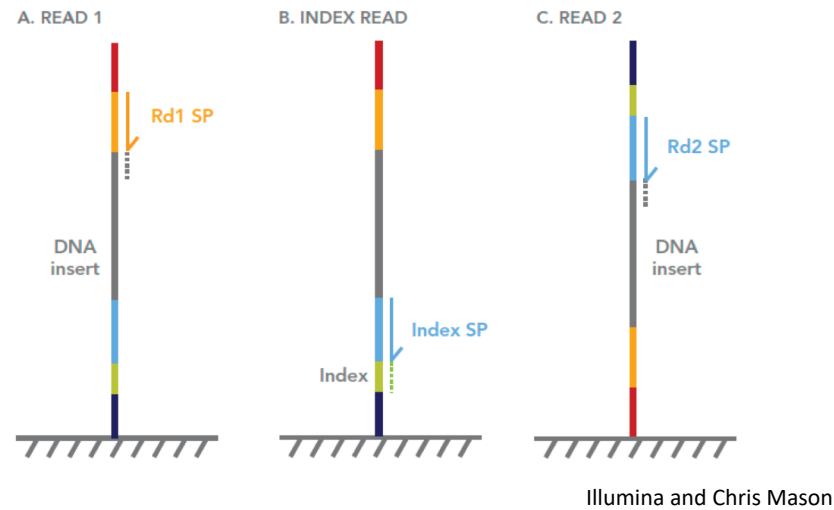
Contig is a stretch of unambiguously assembled sequence

Scaffolding *orders* and *orients* contigs with respect to each other

For this we can use data from various sources, especially *paired ends*

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Indexed sequencing method is now standard for single and paired reads



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Scaffolding: paired-end sequencing

Alternative protocol produces a *pair* of reads taken from either end of a longer *fragment*



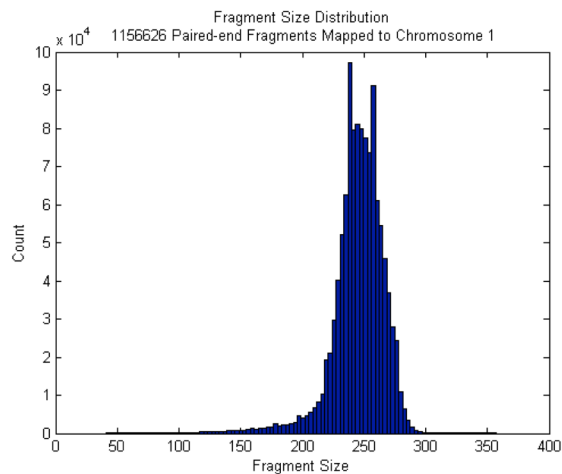
Depending on lengths, mates might overlap in the middle of the fragment

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Scaffolding: paired-end sequencing

Example fragment length distribution

Fragments are not exactly the same length, but there's a clear peak around 250 nt, very few < 150 nt or > 300 nt

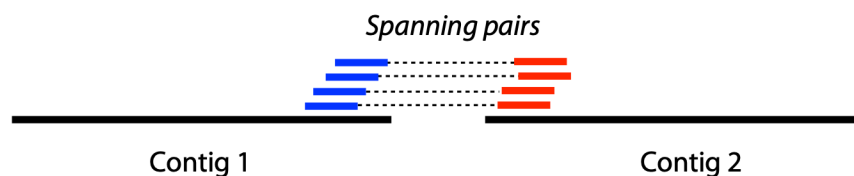


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Scaffolding: paired-end sequencing

Say we have a collection of pairs and we assemble them as usual

Assembly yields two contigs:



...and we discover that some of the mates at one edge of contig 1 are paired with mates in contig 2

Call these *spanning pairs*

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Scaffolding: paired-end sequencing

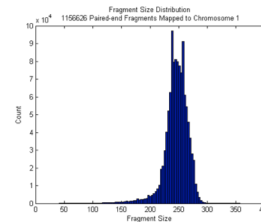


What does this tell us?

Contig 1 is close to contig 2 in the genome

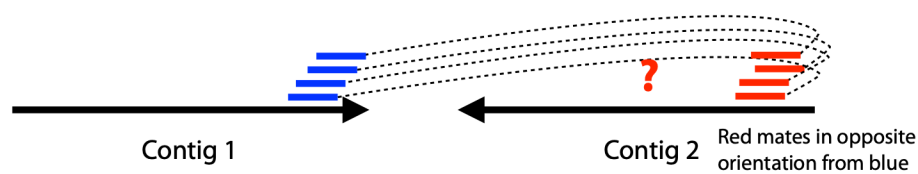
In fact, we can *estimate distance between contigs* using what we know about fragment length distribution

The more spanning pairs we have, the better our estimate

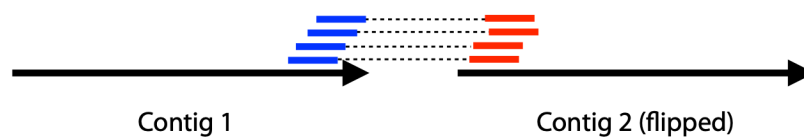


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Scaffolding: paired-end sequencing



What does the picture look like if contigs 1 and 2 are close, but we assembled contig 2 "backwards" (i.e. reverse complemented)

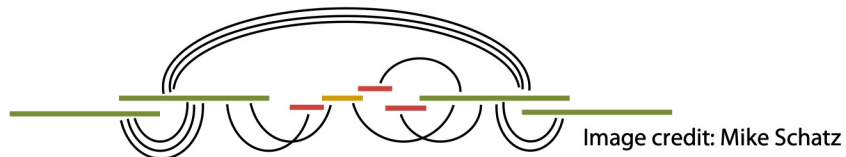


Pairs also tell us about contigs' *relative orientation*

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Scaffolding

Scaffolding output: collection of *scaffolds*, where a scaffold is a collection of contigs related to each other with high confidence using pairs



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Why are genomes hard to assemble?

1. **Biological:**

- (Very) High ploidy, heterozygosity, repeat content

2. **Sequencing:**

- (Very) large genomes, imperfect sequencing

3. **Computational:**

- (Very) Large genomes, complex structure

4. **Accuracy:**

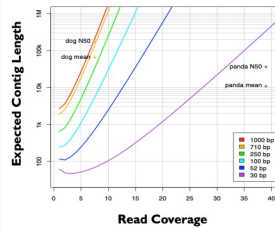
- (Very) Hard to assess correctness



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Ingredients for a good assembly

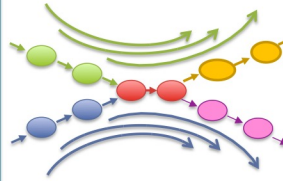
Coverage



High coverage is required

- Oversample the genome to ensure every base is sequenced with long overlaps between reads
- Biased coverage will also fragment assembly

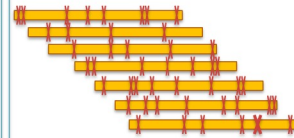
Read Length



Reads & mates must be longer than the repeats

- Short reads will have *false overlaps* forming hairball assembly graphs
- With long enough reads, assemble entire chromosomes into contigs

Quality

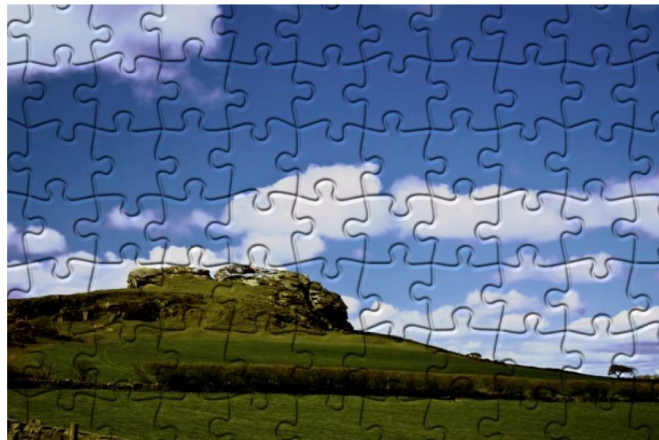


Errors obscure overlaps

- Reads are assembled by finding kmers shared in pair of reads
- High error rate requires very short seeds, increasing complexity and forming assembly hairballs

Current challenges in *de novo* plant genome sequencing and assembly
 Schatz MC, Witkowski, McCombie, WR (2012) *Genome Biology*. 12:243

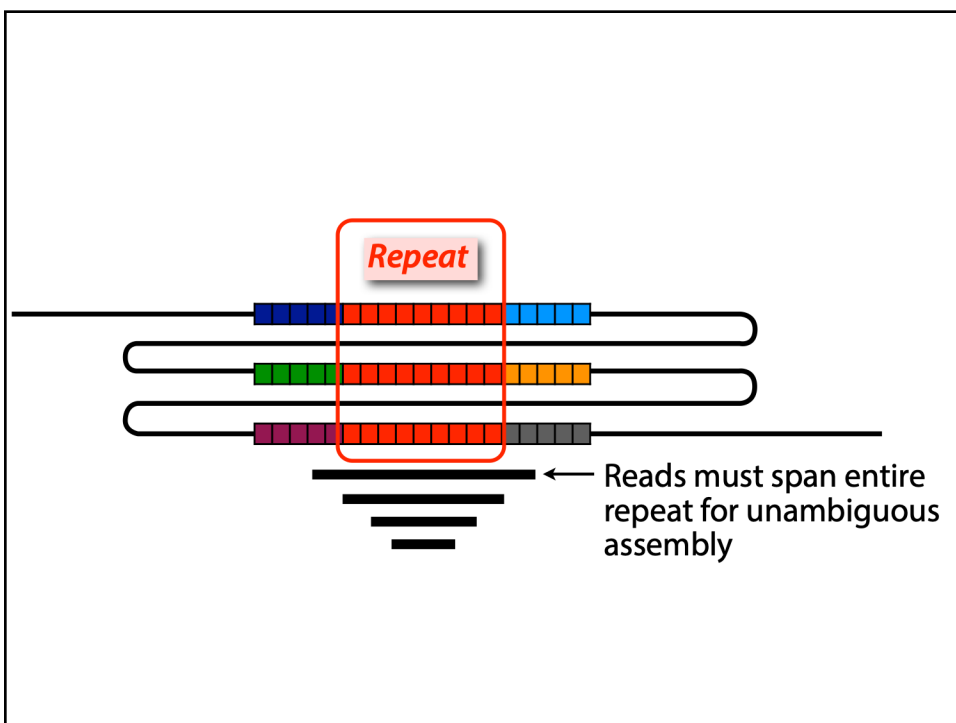
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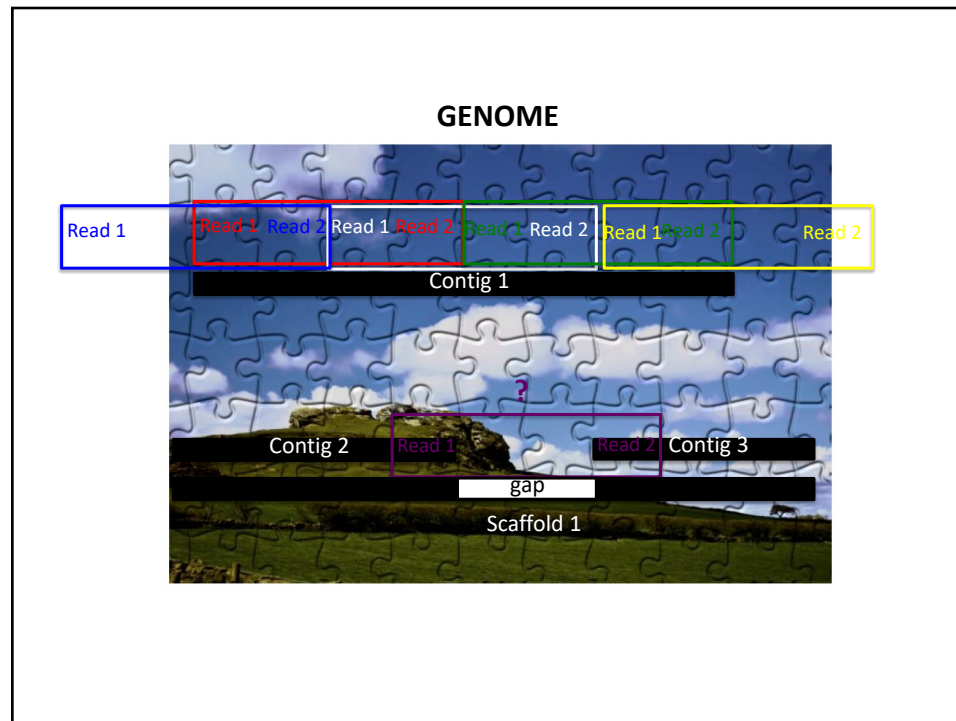
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Running an assembly (SPAdes)

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Pp. 455–477
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Original Articles

SPAdes: A New Genome Assembly Algorithm and Its Applications to Single-Cell Sequencing

ANTON BANKEVICH,^{1,2} SERGEY NURK,^{1,2} DMITRY ANTIPOV,¹ ALEXEY A. GUREVICH,¹
MIKHAIL DVORKIN,¹ ALEXANDER S. KULIKOV,^{1,3} VALERY M. LESIN,¹
SERGEY I. NIKOLENKO,^{1,3} SON PHAM,⁴ ANDREY D. PRJIBELSKI,¹ ALEXEY V. PYSHKIN,¹
ALEXANDER V. SIROTKIN,¹ NIKOLAY VYAHHI,¹ GLENN TESLER,⁵
MAX A. ALEKSEYEV,^{1,6} and PAVEL A. PEVZNER^{1,4}

- Uses modified Debrujin graphs (**a multisized debrujin graph**)
- Very good for assembly of single cell sequencing (amplified genomes)

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```
nohup spades.py -1 Run1-good_1.fastq -2 Run1-  
good_2.fastq --s1 Run1-good_1_singletons.fastq --s2  
Run1-good_2_singletons.fastq -o Run1-spades-assembly  
--only-assembler -m 100 &
```

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Sequencing types

- Whole genome sequencing (human, plant, bacteria, etc.)
- Single cell sequencing
- Metagenomic sequencing
- RNA-Seq
- etc.

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Single cell sequencing

- Genetic variation between different tissues

Extensive genetic variation in somatic human tissues

Maeve O'Huallachain^{a,b}, Konrad J. Karczewski^{a,c}, Sherman M. Weissman^{d,1}, Alexander Eckehart Urban^{a,e}, and Michael P. Snyder^{a,1}

Mosaic Copy Number Variation in Human Neurons

Michael J. McConnell^{1,2,7,8,9}, Michael R. Lindberg⁷, Kristen J. Brennand^{1*}, Julia C. Piper^{1,2,†}, Thierry Voet^{3,4}, Chris Cowing-Zitron⁷, Svetlana Shumilina⁷, Roger S. Lasken^{5,6}, Joris R. Vermeesch³, Ira M. Hall^{7,9,‡}, Fred H. Gage^{2,‡}

- Genetic variation between healthy and diseased cells

Tumour evolution inferred by single-cell sequencing

Nicholas Navin^{1,2}, Jude Kendall¹, Jennifer Troge¹, Peter Andrews¹, Linda Rodgers¹, Jeanne McIndoo¹, Kerry Cook¹, Asya Stepansky¹, Dan Levy¹, Diane Esposito¹, Lakshmi Muthuswamy³, Alex Krasnitz¹, W. Richard McCombie¹, James Hicks¹ & Michael Wigler¹

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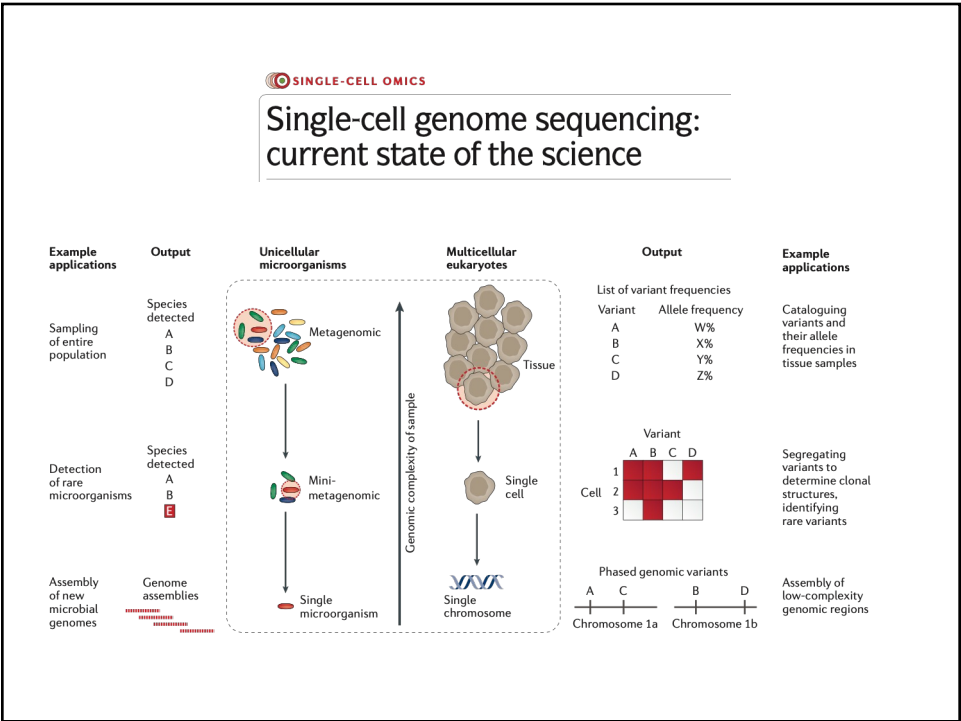
Single cell sequencing

- Sequencing of uncultured organisms

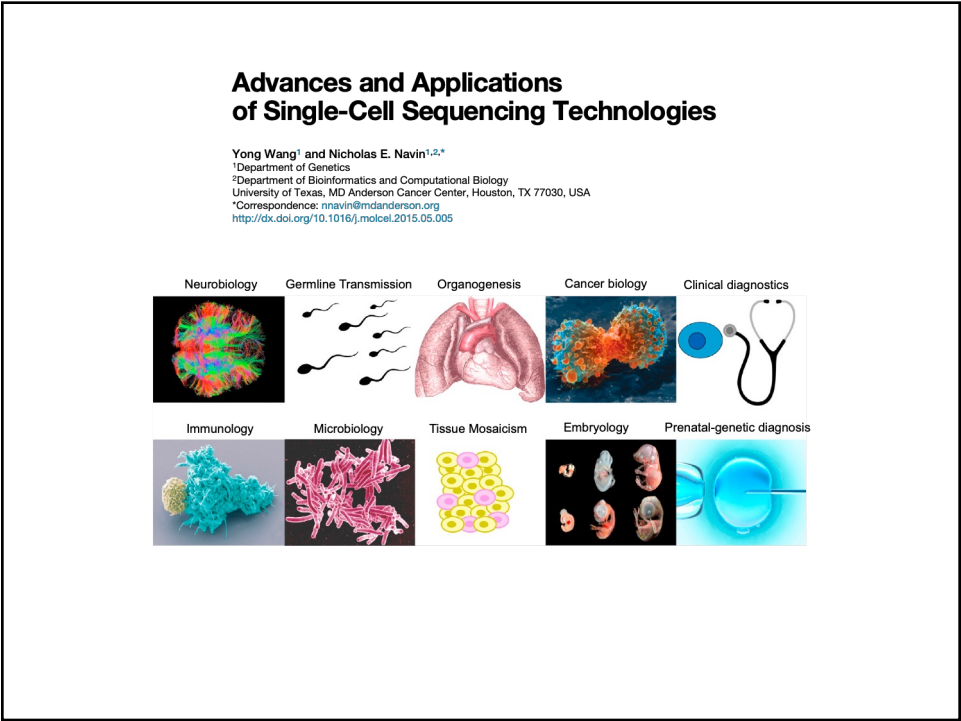
Insights into the phylogeny and coding potential of microbial dark matter

Christian Rinke¹, Patrick Schwientek¹, Alexander Sczyrba^{1,2}, Natalia N. Ivanova¹, Iain J. Anderson^{1,†}, Jan-Fang Cheng¹, Aaron Darling^{3,4}, Stephanie Malfatti¹, Brandon K. Swan⁵, Esther A. Gies⁶, Jeremy A. Dodsworth⁷, Brian P. Hedlund⁷, George Tsiamis⁸, Stefan M. Sievert⁹, Wen-Tso Liu¹⁰, Jonathan A. Eisen³, Steven J. Hallam⁶, Nikos C. Kyrpides¹, Ramunas Stepanauskas⁵, Edward M. Rubin¹, Philip Hugenholtz¹ & Tanja Woyke¹

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How does it work?

What are the biggest hurdles in Single Cell Sequencing?

- Cell isolation
- Whole Genome Amplification
- Data analysis